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Spring 26 BAN 650

Assignment 8

Assignment Open Date: April 8, 2026

Assignment Due Date: April 22, 2026

Task 1:

SLA Breach Rate by System and Priority

SLA targets assumed: P1=8h, P2=16h, P3=24h, P4=48h

Breach Rate

0.00%  86.30%

System	Priority			
	P1	P2	P3	P4
CRM	86.10%	67.15%	52.46%	0.00%
Email	81.71%	66.37%	50.00%	0.00%
ERP	84.56%	69.21%	50.52%	0.00%
Network	86.30%	66.45%	50.77%	0.00%

Does it clearly support a decision or action?

- Yes — the viz points directly to P1 incidents as the area needing management attention
- The uniform 82–86% breach rate across all four systems rules out a system-specific fix
- Directs the manager toward capacity, staffing, or process changes for high-severity tickets rather than tooling investments

Could it be misinterpreted or over-trusted?

- Yes — the clean red/green color coding makes the breach rate look like an audited, official figure
- A manager could accept 86% at face value and escalate without questioning the SLA thresholds used to calculate it
- The strong visual contrast may also draw attention away from P2 and P3 tiers, which still have 50–69% breach rates worth addressing

Are assumptions or limitations visible?

- Partially — the subtitle states "SLA targets assumed: P1=8h, P2=16h, P3=24h, P4=48h"
- However, the viz does not explain *why* those thresholds were chosen or that they came from general ITIL conventions rather than the organization's actual SLA policy
- Sample sizes (ticket counts per cell) are not shown, so the reader can't judge statistical reliability
- State-level, issue-type, and time-based variations are hidden by aggregation

Identified Risk — Automation Bias layered on Hidden Assumptions

- The most serious risk is automation bias: the viz looks polished and definitive, so a manager may trust it without questioning the inputs
- The SLA thresholds are my defaults, not values validated with the service-desk owner
- If the actual organizational SLA targets differ, the breach rates shown could be materially wrong
- This risk is compounded by over-aggregation — individual analyst, state, and issue-type patterns are invisible, so any decision based on this view alone would be premature
- Operational performance data is also sensitive — aggregated views like this could be drilled down to identify specific analysts or locations, raising HR and privacy concerns if shared broadly

What insight the visualization provides to a manager

- Reveals that P1 (highest-severity) incidents breach SLA in 82–86% of cases across all four systems (CRM, ERP, Email, Network)
- Shows breach rates decline sharply by priority tier: P1 ~85%, P2 ~67%, P3 ~51%, P4 ~0%
- Uniform pattern across all four systems rules out a single-system problem
- Points to urgent-ticket response capacity as the binding operational constraint
- Suggests that tooling or staffing investments targeted at one specific system would not address the shortfall

Task 2

How AI-assisted features were used and why

- Used Claude (AI assistant) to brainstorm which operational angle would be most meaningful; it suggested framing resolution time against standard ITIL-style SLA thresholds (P1=8h, P2=16h, P3=24h, P4=48h) to convert raw hours into a breach rate
- Claude also recommended the heatmap layout (System × Priority) as the clearest way to surface both magnitude and concentration at a glance
- Tableau's Show Me panel independently confirmed the heatmap as the best fit for two categorical dimensions and one measure
- Built the calculated field in Tableau to flag each ticket as a breach, then used AVG over that flag to compute the breach rate per cell
- Applied a red-to-green color ramp so the eye is drawn immediately to the P1 column where management attention is needed
- AI was used as a thinking partner for framing and design choice — all formulas, calculations, and the final viz were built and validated in Tableau

Risks, ethical concerns, and governance considerations

- **Automation bias:** the viz looks polished and definitive, so a manager may trust the 86% figure as an audited fact without questioning the inputs
- **Hidden assumptions:** SLA thresholds are defaults I applied, not values validated with the service-desk owner; if actual organizational SLAs differ, the breach rates shown could be materially wrong
- **Over-aggregation:** state-level, issue-type, analyst-level, and time-based variation are invisible, so any decision based on this view alone would be premature
- **Sensitivity of operational data:** aggregated performance data could be drilled down to identify specific analysts or locations, raising HR and privacy concerns if shared broadly
- **Governance gap:** no documentation of data lineage, calculation logic, or threshold rationale accompanies the viz

How a manager should interpret the visualization responsibly

- Treat the viz as a prompt for investigation, not a verdict
- Validate the SLA thresholds with the service-desk owner before acting on the breach percentages
- Drill into lower-level breakdowns (by state, issue type, time period, analyst) before assigning cause
- Consider whether high-severity tickets have genuinely tighter response requirements or whether thresholds need re-baselining
- Pair this view with qualitative context (staffing levels, recent outages, process changes) before making capacity or budget decisions
- Limit circulation of detailed breakdowns to audiences with a legitimate need, given the sensitivity of performance data